

MobiDock, a Norwegian company, is putting 400 charging stations outside in a city of 15,000 people for electric two- and three-wheelers.

The infrastructure of shared micro-mobility provides a two-wheeler to go up to a nearby bus or metro station. The same micro-mobility can be used, after alighting from the bus or metro, to go to the final destination. As everything is connected to the cloud, the

user can use the same ticket for the whole journey.

For seamless connectivity, every electric kick bike is equipped with an IoT charging device that automatically handshakes with the charging station and commences charging if they have an IoT match (that is both parties have an agreement to charge). The IoT charging device also equips a DC power charger, thus combining connectivity, data transfer, and charging in the same technology.

The technology for these applications has a data link that communicates with the cloud and allows a bike to be taken on rent. Once rented, the user is notified on the phone and everything from there is seamless. The user can view activities such as battery status, position of the bike, and available parking station. On a larger scale, shared mobility is more advantageous for private rental bike companies to enter the market without too much barrier in terms of infrastructure.

Similar to rental bikes, private bikes can also use the same infrastructure where the consumer is charged every time they stop. The usual startup cost and an hourly charge after being parked continues to remain the same. The ticket for a month for these bikes can be taken as a combination with other modes of transport such as metro and bus for a complete month.

Other micro-mobility applications

Micro-mobility can also be applied in aerial drones. The drones can

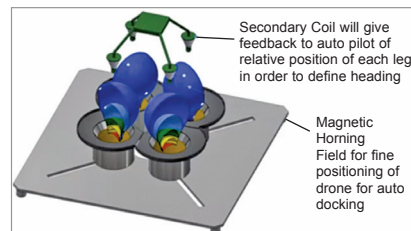


Fig. 7: Drone docked for charging of its battery

be charged through their feet at their parking lot by inducing magnetic field. The secondary coil in a drone gives information about relative position of its each leg to the autopilot to define heading. The primary coil in the parking station induces a magnetic field for fine positioning of the drone for automatic docking. The same technology can be used for charging boats on ports as well.

The autonomous robots can be used for agricultural applications too to enhance farm yields. Agriculture needs a lot of automation that can be handled through robots and tractors, and now in combination with aerial drones. This mobility enables moving things autonomously through a robust wireless, rather than moving people.

Micro-mobility in India

In a country like India, 2-wheelers are in big demand and micro-mobility is growing by a good margin as well. Statistics suggest 66% of the traffic in India is micro-mobile.

But the challenges that we face are currently global. The need for highly efficient and low-cost charging devices for electric vehicles is felt all over the world.

Helge Sverre Eide, Partner and Managing Director, Wireless Power and Communication (WPC Norway) says, "The government policy on electric vehicles is encouraging as it gives incentives to those who actually move into this EV business. Norway has utilised this procedure to promote EVs and ban the sale of combustion cars in the future. The government has taken a lot of effort for this, and I feel that India is doing the same." **EFY**

This article, based on a presentation by Helge Sverre Eide, Partner and Managing Director, Wireless Power and Communication (WPC Norway) at MOVES 2021, was prepared by Darshil Patel, a technology enthusiast with a deep interest in power electronics.

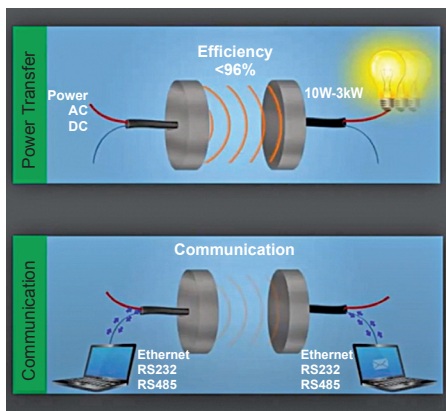


Fig. 4: Wireless efficiency and communication between primary and secondary connectors

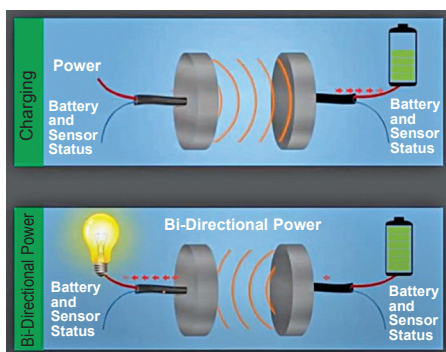


Fig. 5: Wireless charging and bi-directional power between primary and secondary connectors

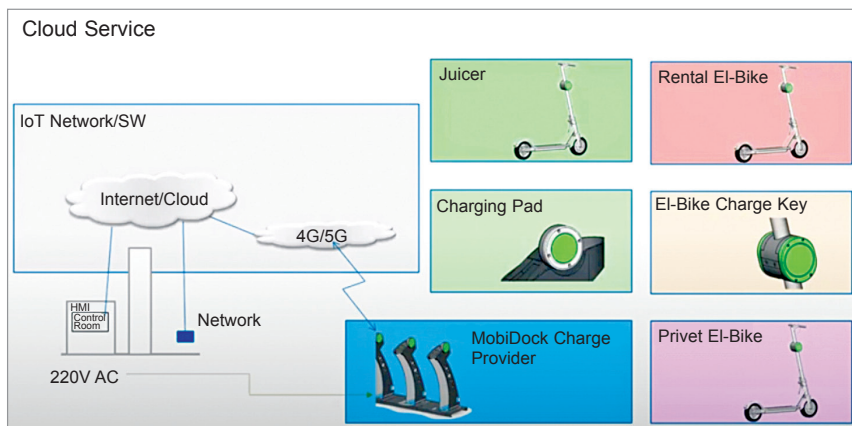


Fig. 6: Connectivity between the IoT network, charger, and the device